WELLALIGN

Middlesex University Mauritius

UG- Final Year Project

AI | Posture | Computer Vision | Machine Learning | Physical Wellbeing

AI-POWERED POSTURE CORRECTION AND EXERCISE RECOMMENDATION APPLICATION FOR PHYSICAL WELLBEING

Problem Statement :

Postural health is often overlooked in the society. Poor posture in desk-based workers and students causes musculoskeletal strain and reduced wellbeing. Existing tools focus only on classification performance or provide limited real-world guidance. There is a gap between isolated posture predictions and end-to-end support workflows that detect issues, explain findings, recommend actions, and enable longitudinal progress tracking.

Key Features :

✓ **Image-based Assessment**: upload front/side posture photos → ML analysis → scoring
✓ **Real-time Monitoring**: live webcam tracking → issue detection → session summaries
✓ **AI Prescription**: Gemini-powered guidance → exercise recommendations → video links
✓ **Progress Tracking**: historical assessments/sessions → trend charts → personalisation
✓ **Safety-First**: medical disclaimers, non-diagnostic framing

Evaluation :

Model Performance:

- Validation Accuracy: 0.98 | Macro F1: 0.97 | Weighted F1: 0.98

System Testing:

- End-to-end workflows reliable under normal and failure conditions

Feature Utility:

- AI enrichment improves feedback interpretability and actionability
- Historical persistence enables longitudinal trend analysis and personalisation

Technology Used:

Frontend: HTML, CSS, JS, Vue.js, Tailwind CSS

Backend: node JS, Express, MongoDB

External APIs: Gemini, YouTube Data API v3

ML Service: scikit learn, python, OpenCV, MediaPipe

Project Author & Supervisor :

Author : Dhruv Ramachundrah
StudentID : M01012616
Supervisor: Haardik Gautum

Contact me on: ramachundrahdhruv@gmail.com